


# AI Voice-Controlled ERP Assistant

Natural Language Interface for ERP Operations — a system that enables users to interact with enterprise resource planning functions using voice or text commands, eliminating the need to manually navigate complex ERP screens and forms.

# Problem Statement & Objective

## The Problem

Enterprise ERP systems are powerful — but notoriously difficult to use. Accessing even routine data requires navigating multiple screens, filling forms, and understanding complex menu hierarchies.

- Traditional ERP navigation is slow, multi-step, and error-prone
- Repetitive operations consume disproportionate time for routine tasks
- Non-technical users face steep learning curves with complex interfaces

## Our Objective

**Enable users to interact with ERP functions using natural language — voice or text — instead of manually navigating the system.**

## The Solution Architecture

### Voice / Text Input

User speaks or types a command in plain language

### NLP Processing

Intent and entity extraction identify what the user wants

### ERP Operation

Conversational interface with clarification and confirmation executes the task

# What Our System Does

## Core ERP Capabilities

### Inventory Management

Query stock levels by item code or name — e.g., *"Show stock for item P-102"*

### Attendance Operations

Mark and retrieve attendance with write confirmation — e.g., *"Mark my attendance"*

### Sales Reports

Retrieve aggregated sales data and summaries — e.g., *"Get total sales from last month"*

### Employee Operations

Access employee-related information and records on demand

## Interaction Features

### Dual Input

Voice and text input supported interchangeably

### Intent + Entity

Understands *what* to do and *which* records to act on

### Auto-Clarification

Prompts user when required information is missing

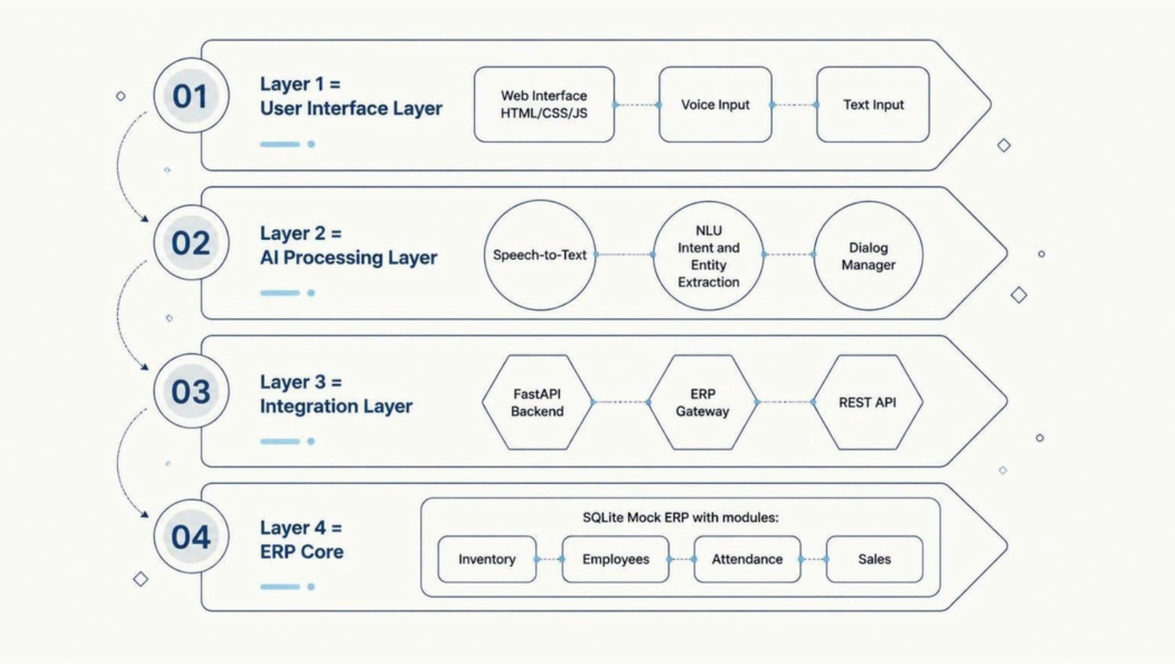
### Write Guard

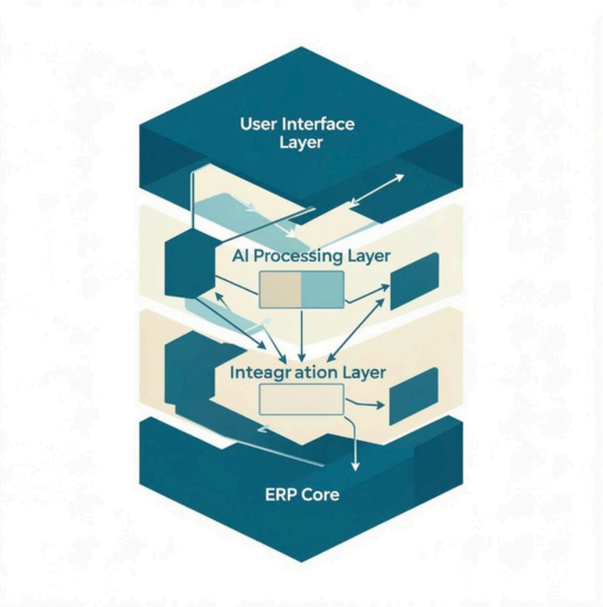
Confirmation required before any write operation executes

> **Real examples:** "Show stock for item P-102" · "Mark my attendance" · "Get total sales from last month"

# System Architecture

The system is organized into four distinct layers, each with a clear responsibility — from capturing user input to executing ERP operations on the database.




## UI Layer

Web-based interface accepting both voice and text input. Provides real-time feedback and displays assistant responses.

## AI Processing Layer

Core intelligence: transcribes speech, extracts intent and entities, and manages multi-turn conversation state.

## Integration Layer

FastAPI backend routes validated requests through the ERP Gateway to the appropriate ERP function via REST.

## ERP Core

SQLite mock ERP stores Inventory, Employees, Attendance, and Sales data — mirroring a real ERP schema.

# End-to-End Request Flow

Example walkthrough: ***"Show stock for item P-102"*** — tracing a command from spoken input to delivered response.

1

### ① Voice / Text Input

User speaks or types the command into the web interface

2

### ② Speech-to-Text

Audio converted to text via Whisper or Vosk — text input bypasses this step and enters NLU directly

3

### ③ NLU Processing

Intent = **Stock Query** | Entity = **P-102** — extracted via rule-based or LLM-based NLU

4

### ④ Dialog Manager

Validates completeness and intent reliability; requests clarification or confirmation if needed

5

### ⑤ ERP Gateway

Routes validated request to the appropriate ERP function via FastAPI

6

### ⑥ Database Query

SQLite retrieves stock information for item P-102

7

### ⑦ Response

Result formatted as text and optionally converted to speech via pyttsx3

Text input bypasses Speech-to-Text and enters the processing pipeline directly at the NLU stage — enabling fast, low-latency interactions without audio hardware.

# NLU, Entity Resolution & Dialog Management

## 1. Natural Language Understanding (NLU)

1

### Intent Recognition

Determines *what* the user wants — e.g., Stock Query, Attendance Mark, Sales Report

2

### Entity Extraction

Identifies relevant objects — item codes, employee IDs, date ranges

3

### Dual NLU Modes

**Rule-based** (default): fast, deterministic, pattern-limited via regex
**LLM-based** (optional): Ollama + Llama 3.2, more flexible, handles natural variation

## 2. Entity Resolution

Users don't need to know exact ERP record IDs. Fuzzy matching connects spoken or name-based references to ERP records — *"hydraulic valve"* resolves to **P-102**. Uncertain matches trigger a clarification prompt rather than silent, potentially incorrect guessing.

## 3. Dialog Manager — Conversation Intelligence

The Dialog Manager is the safety and intelligence layer that ensures operations are complete, confirmed, and correct before execution.

### Missing Information Flow

User: *"Check stock"*

Assistant: *"Which item would you like to check?"*

User: *"P-101"* → Result returned

### Write Confirmation Flow

User: *"Mark my attendance"*

Assistant: *"Confirm marking attendance for today?"*

User: *"Yes"* → Operation executed

Ensures **data integrity** and **user intent clarity** before any database write.

# Technology Stack



## Frontend

- HTML, CSS, JavaScript
- Web-based UI with dual voice and text input
- Real-time response display



## Backend & Processing

- Python, FastAPI, Uvicorn, Pydantic
- Rule-based NLU via regex/pattern matching
- Optional: Ollama + Llama 3.2 for LLM-based NLU



## Speech Processing

- Speech-to-Text: Whisper (API/local) or Vosk
- Text-to-Speech: pyttsx3
- Pluggable architecture — swap STT/TTS engines independently



## Data & Communication

- Database: SQLite (mock ERP schema)
- API: HTTP/REST communication
- Modules: Inventory, Employees, Attendance, Sales

# Key Demonstrated Use Cases

The following use cases are fully implemented and demonstrable end-to-end in the current system.

### 📦 Inventory Query

> "Show stock for item P-102"

System resolves item code, queries SQLite inventory table, and returns current stock level in a formatted response.

### 🗓️ Attendance Management

> "Mark my attendance"

Dialog Manager triggers confirmation prompt. On user approval, attendance record is written to the database — no accidental writes.

### 📊 Sales Report

> "Get total sales from last month"

Date entity extracted, ERP Gateway queries the sales table, aggregated total returned and spoken aloud via TTS.

### 🔄 Clarification Flow

> "Check stock" → "Which item?" → "P-101"

Incomplete commands trigger a targeted follow-up. System awaits the missing entity before routing to ERP — demonstrating multi-turn dialog.

**Interactive Mode:** Users can enter custom commands and observe the system's intent extraction, entity resolution, dialog state, and ERP response in real-time during the demonstration.

# Current Implementation & Future Scope

## Currently Implemented ✓

### End-to-End Text Pipeline

Working input-to-response pipeline via web interface and FastAPI backend

### Mock SQLite ERP

Inventory, Attendance, Sales, and Employees modules with realistic schema

### Rule-Based NLU

Intent recognition and entity extraction via regex/pattern matching

### Dialog Manager

Clarification loops and write confirmation before all database mutations

### Fuzzy Entity Resolution

Name-based references matched to ERP records without exact ID input

### Pluggable Speech I/O

Whisper/Vosk for STT · pyttsx3 for TTS — swappable without core changes

## Limitations & Future Enhancements →

### NLU Flexibility

Rule-based NLU has limited language variation coverage — enhance with optional Ollama LLM integration already scaffolded

### Real ERP Integration

Replace mock SQLite with production ERP — Odoo, SAP, or PostgreSQL-backed systems

### Production Security

Implement OAuth2/JWT authentication (currently placeholder) + rate limiting and security hardening

### Multilingual Support

Extend NLU and STT to support regional and international languages

### Proactive Alerts

ERP-driven notifications — low stock alerts, attendance anomalies, sales thresholds

### Cloud Deployment

Docker containerization and cloud deployment for scalable, production-ready operation

# Conclusion

The AI Voice-Controlled ERP Assistant demonstrates a working natural-language interface for ERP operations — connecting voice and text input through NLU, dialog management, and API integration to deliver ERP data and actions without manual navigation.

1

### User Input

Voice or text command

2

### NLU

Intent + entity extraction

3

### Dialog Manager

Clarify + confirm

4

### ERP Gateway

Route + execute

5

### Response

Text + speech output

### Simplifies ERP Interaction

Natural language replaces complex multi-screen navigation — any user can access ERP functions without training

### Flexible Input Methods

Voice and text workflows supported interchangeably — the same pipeline handles both modalities seamlessly

### Foundation for Integration

Architecture is designed for real ERP systems — swap SQLite for Odoo or PostgreSQL and deploy to production

## Thank You

We welcome your questions and feedback.